Day 16 Task: HTTP and HTTPS

HTTP (Hypertext Transfer Protocol) is an application-layer protocol used for communication between web browsers and web servers. It allows clients to request web pages, images, videos, and other resources from servers.

HTTPS (Hypertext Transfer Protocol Secure) is the secure version of HTTP. It uses Transport Layer Security (TLS) to protect communication between a client and a server. HTTPS helps prevent attackers from reading or modifying data exchanged over the network.

Secure web communication is essential for protecting passwords, personal information, financial transactions, and other sensitive data.

What Is HTTP?

HTTP is a protocol that defines how clients and servers exchange information on the web. When a user opens a website, the browser sends an HTTP request to the server, and the server responds with the requested resource or an appropriate status code.

HTTP does not provide encryption by itself. Therefore, information transmitted over ordinary HTTP can potentially be intercepted or modified by an attacker who can observe or interfere with the network traffic.

Default port: TCP port 80.

What Is HTTPS?

HTTPS is HTTP communication protected by TLS. It establishes a secure connection between the client and the server, helping protect the confidentiality and integrity of data in transit.

HTTPS also uses digital certificates to authenticate the server's identity. This helps users verify that they are communicating with the intended website, provided the certificate is valid and correctly verified.

Default port: TCP port 443 for HTTP/1.1 and HTTP/2 over TLS. HTTP/3 uses QUIC over UDP, typically on port 443.

Difference Between HTTP and HTTPS

Feature	HTTP	HTTPS
Full form	Hypertext Transfer Protocol	Hypertext Transfer Protocol Secure
Security	No encryption by default	Uses TLS to protect communication
Encryption	Not provided by HTTP itself	Encrypts data in transit
Authentication	No TLS certificate authentication	Can authenticate the server using a digital certificate
Data integrity	No built-in cryptographic protection	TLS helps detect unauthorized modification
Default port	80	443
URL prefix	http://	https://
Sensitive information	Unsafe to transmit without additional protection	Appropriate when correctly configured
Common use	Unencrypted web communication	Login pages, banking, shopping, and modern websites

Security Benefits of TLS

TLS stands for Transport Layer Security. It is a cryptographic protocol that protects communication between networked applications.

Encryption

TLS encrypts transmitted data so that unauthorized parties who intercept the traffic cannot readily read its contents. This helps protect passwords, messages, and personal information.

Authentication

TLS can authenticate a server using a digital certificate issued by a trusted Certificate Authority. The browser checks the certificate and the website's domain name to help confirm the server's identity.

Data Integrity

TLS uses cryptographic protection to detect unauthorized changes to data while it is being transmitted. This helps ensure that information arrives without undetected tampering.

Protection Against Network Attacks

When correctly configured and used, HTTPS reduces the risk of eavesdropping, data manipulation, and certain man-in-the-middle attacks. However, it cannot prevent every cyberattack, such as phishing or malware on a compromised device.

How HTTPS Works

The HTTPS communication process involves the following steps:

- Connection request: The browser connects to the web server using HTTPS.
- TLS handshake: The browser and server negotiate security parameters and establish cryptographic keys.
- Certificate verification: The browser checks the server's certificate and whether it matches the requested domain.
- Secure connection: Once the handshake succeeds, the parties use cryptographic keys to protect application data.
- Data exchange: HTTP requests and responses travel through the protected TLS connection.

This process helps ensure that communication is encrypted, authenticated, and protected against undetected modification.

Why HTTP Is Insecure for Sensitive Data

HTTP does not encrypt information by itself. If sensitive data is sent over an unprotected HTTP connection, an attacker with access to the network traffic may be able to read or alter it.

For example, if a user submits login credentials through an ordinary HTTP connection, those credentials may be exposed to interception. Attackers may also modify web content or redirect users to malicious destinations.

HTTPS reduces these risks by protecting data in transit with TLS. Therefore, websites should use HTTPS throughout the site, especially for authentication, payments, and personal information.

Limitations of HTTPS

Although HTTPS is essential for secure web communication, it does not guarantee that a website is legitimate or harmless. Attackers can also obtain valid TLS certificates for malicious websites.

HTTPS does not automatically protect a device from malware, prevent phishing, or secure data stored on a server. Website security also requires secure coding, strong authentication, regular updates, and appropriate server-side protections.